

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech Branch : Electical Engineering / Electrical Engg (Electronics & Power)/Electrical & Electronics Engg/Electrical & Power Engineering

Semester :VII

Subject Code & Name: BTEOE704B & Biomedical Instrumentation

Max Marks: 60

Date: 12/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The variation of the electrical potential associated with the passage of a pulse along the membrane of a muscle cell or a nerve cell is called _____ a. muscle potential b. action potential c. Resting potential d. half-cell potential		1
2	Phonocardiography is listening to _____ a. arm muscle sound b. lungs sound c.Heart sound d. Respiratory sound		1
3	Recording electrical activities associated with human brain is known as _____ a. EEG b. ECG c. EMG d. EOG		1
4	Which of the following is considered to be the primary pacemaker of the heart? a. sino-atrial node b. atrio-ventricular node c. purkinje fibres d. bundle of his		1
5	_____ are devices which convert one form of energy into another. a. Transducers b. electrode c. Op-amp d. Impulse		1

6	Which of the following has been used for temperature measurement?					1
	a. RTD	b. Thermo-couple	c. Thermistor	d. all of above		
7	A pulse oximeter shows the percentage of _____					1
	a. venous haemoglobin	b. Saturated O ₂	c. Saturated CO ₂	d. Sodium ions in cell		
8	Transcutaneous means _____					1
	a. passing to the skin	b. passing to the bones	c. passing to the heart	d. passing to the lungs		
9	Blood vessels are _____					1
	a. hard tissue	b. soft tissue	c. connective and hard tissue	d. connective and soft tissue		
10	Which of the following is not the property of the Instrumentational amplifier?					1
	a. Extremely high input impedance.	b. Low bias and offset currents	c. High slew rate	d. Very low CMRR		
11	CMRR is measured in _____					1
	a. Volts/second	b. Decibels	c. Decibels/V	d. Decibels/m		
12	_____ are commonly used for providing protection against leakage currents.					1
	a. Isolation amplifiers	b. Pre-amplifiers	c. Driver amplifiers	d. Power amplifiers		
Q. 2 Solve the following.						12
A)	What are bioelectric potentials? Discuss the origin frequency and voltage of ECG, EMG and EEG signals.				L	6

B)	Explain the measurement of PO ₂ and PCO ₂ with neat construction diagram of electrodes.	U	6
Q.3	Solve the following.		12
A)	Explain pulmonary blood circulation with neat diagram of heart.	U	6
B)	What do you mean by BP measurement? Discuss any one technique of BP measurement.	L	6
Q. 4	Solve Any Two of the following.		12
A)	Draw and schematic diagram of neuron and explain how neurotransmission occurs.	L	6
B)	Explain different types of electrodes used for ECG measurement.	A	6
C)	Explain how spirometer used for respiratory measurements with neat diagram.	A	6
Q.5	Solve Any Two of the following.		12
A)	What are different bio-potentials generated across human cell? Explain with schematic diagram.	L	6
B)	Explain significance of pH measurement.	L	6
C)	What different types of non-electrical parameters in human body? Explain heart rate measurement.	U	6
Q. 6	Solve Any Two of the following.		12
A)	Explain defibrillator with neat block diagram.	L	6
B)	Discuss differences between radiographic and fluoroscopic techniques.	U	6
C)	Explain ventilator with neat block diagram.	U	6
*** End ***			